Heuristics for the structural design of peptides using contact map

Jazmín Alejandra Martínez Guerrero
martinez.ja@cicese.edu.mx
Center for Scientific Research and Higher Education at Ensenada (CICESE)
Ensenada, Baja California, México

ABSTRACT

In this study, heuristic optimization approaches for peptide structural design using contact map surrogates were explored and evaluated, focusing on their ability to guide the search toward structurally accurate three-dimensional conformations. Contact maps predicted by the ESM-2 language model were used to define the optimization objective. Two fitness evaluation strategies based on contact probability matrices were considered and compared: the Area Under the Precision–Recall Curve (AUC-PR) and Shannon entropy.

The sequences generated through this optimization process were subsequently validated using AlphaFold2 to obtain their three-dimensional structures, enabling the computation of structural metrics such as dRMSD and GDT-TS. This allowed a comparative evaluation of both heuristics against a random search baseline, as well as a statistical analysis of the extent to which the fitness functions reflect the structural quality of the resulting conformations.

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1 INTRODUCTION

In this study, heuristic optimization approaches for peptide structural design using contact map surrogates were explored and evaluated, focusing on their ability to guide the search toward structurally accurate three-dimensional conformations. Contact maps predicted by the ESM-2 language model were used to define the optimization objective. Two fitness evaluation strategies based on contact probability matrices were considered and compared: the Area Under the Precision–Recall Curve (AUC-PR) and Shannon entropy.

The sequences generated through this optimization process were subsequently validated using AlphaFold2 to obtain their three-dimensional structures, enabling the computation of structural metrics such as dRMSD and GDT-TS. This allowed a comparative evaluation of both heuristics against a random search baseline, as well as a statistical analysis of the extent to which the fitness functions reflect the structural quality of the resulting conformations.

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1.1 Peptides

When we observe a cell under a microscope or analyze its electrical or biochemical activity, we are essentially observing the work of peptides. Peptides are the main building blocks from which cells are assembled and constitute most of the cell's dry mass.

Some examples of the general functions of peptides are:

- Catalyzing the breaking or formation of covalent bonds.
- Providing mechanical support to cells and tissues.
- Transporting small molecules.
- Generating movement in cells and tissues.
- Regulating transcription

A peptide molecule is formed by a chain of amino acids linked together by covalent peptide bonds. In each type of peptide, the amino acids are arranged in a unique order, called the amino acid sequence, which is exactly the same from one molecule of that peptide to another. In essence, the function of a peptide depends on its three-dimensional structure, which is determined by its amino acid sequence. Therefore, the objective is to identify amino acid sequences that fold into a specific three-dimensional structure capable of performing a defined biological function in a stable and efficient manner [1].

1.2 Peptide Design

Peptide design is the process of creating new peptides with specific properties by manipulating the amino acid sequence. It involves a combination of computational techniques, laboratory experiments, and interdisciplinary knowledge [8].

Some applications of peptide design are [6]:

- The development of enzymes useful in drug production, toxin degradation, and biofuel generation.
- The creation of sensors, such as calcium sensors, used to detect neuronal activity.
- The generation of peptides capable of identifying TNT or other explosives, with applications in mine detection.

1.3 Motivation

In 2024, Leyva et al. introduced Key-Cutting Machine (KCM), an iterative optimization platform for the de novo design of proteins and peptides, whose operation is based on the analogy of a key-cutting machine: it uses a desired backbone geometry as the reference "key" and progressively refines amino acid sequences until their predicted structure fits that template. The algorithm operates through a cyclic process in which it generates sequences, predicts their three-dimensional structure (using ESMFold), evaluates their quality through an objective function, and uses the best sequences to update the generation probabilities in the next iteration.

However, the main bottleneck of KCM is the prediction of the three-dimensional structure of the sequences. Although the system

is parallelizable, in its implementation all sequences depend on a single instance of ESMFold running sequentially on a single GPU. This prediction time increases with peptide size; while sequences of up to 100 residues take less than 2 seconds to predict, those with 400 residues require approximately 10 times more computation time [13].

On the other hand, one possible way to represent the three-dimensional structure of a protein is through contact maps. Contact maps essentially indicate whether two residues are in contact with each other. Two amino acids are considered to be in contact when the distance between certain atoms (typically alpha or beta carbons) is below a given threshold (usually < 8 Ångströms) [16].

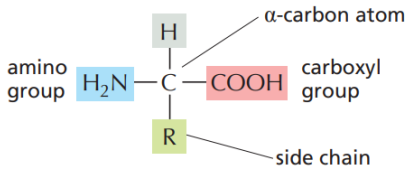
Predicting a contact map is computationally less expensive than predicting the full 3D structure. Although it does not contain the same level of information as a complete structural prediction, it could serve as a sufficiently robust representation to guide the search for sequences with a specific folding. Therefore, **the objective of this project is to explore and evaluate heuristic optimization approaches for peptide structural design using contact map surrogates, focusing on their ability to guide the search toward structurally accurate three-dimensional conformations**.

2 PREVIOUS CONCEPTS

2.1 Amino Acids

Amino acids are small organic molecules with a defining property: they all possess a carboxylic acid group and an amino group, both attached to a central α carbon atom. This α carbon also carries a specific side chain, whose identity distinguishes one amino acid from another [1]. When amino acids are linked through peptide bonds to form a peptide, each loses a water molecule and becomes referred to as a residue; this term refers to the portion of the amino acid that remains integrated into the polypeptide chain.

Figure 1: General structure of an amino acid



All organisms use the same set of 20 amino acids to make their own proteins [1].

2.1.1 Mutations in Amino Acids. Amino acid mutations in peptides consist of the substitution of one residue for another within the peptide sequence. These substitutions can lead to loss or gain of function, structural instability, or abnormal folding [4]. Substitutions that drastically change size or hydrophobicity in the interior of the peptide can destabilize its internal structure, distorting the binding site or even the overall structure [3]. Some studies, in order to evaluate the efficiency of structural predictors, have identified point mutations aimed at maximizing alterations in peptide structures [7], such as those shown in the table1.

Figure 2: Set of the 20 essential amino acids

AMINO ACID			SIDE CHAIN
Aspartic acid	Asp	D	negatively charged
Glutamic acid	Glu	E	negatively charged
Arginine	Arg	R	positively charged
Lysine	Lys	K	positively charged
Histidine	His	H	positively charged
Asparagine	Asn	N	uncharged polar
Glutamine	Gln	Q	uncharged polar
Serine	Ser	S	uncharged polar
Threonine	Thr	T	uncharged polar
Tyrosine	Tyr	Y	uncharged polar
POLAR AMINO ACIDS			

AMINO ACID			SIDE CHAIN
Alanine	Ala	A	nonpolar
Glycine	Gly	G	nonpolar
Valine	Val	V	nonpolar
Leucine	Leu	L	nonpolar
Isoleucine	Ile	I	nonpolar
Proline	Pro	P	nonpolar
Phenylalanine	Phe	F	nonpolar
Methionine	Met	M	nonpolar
Tryptophan	Trp	W	nonpolar
Cysteine	Cys	C	nonpolar
NONPOLAR AMINO ACIDS			

Table 1: Point mutation transformation pairs used for maximal disruption to protein structures. Standard abbreviations are used for each of the 20 canonical amino acids.

From	To	From	To	From	To	From	To
S	W	N	A	K	G	F	T
T	F	Q	G	R	A	M	Q
C	W	Y	G	W	S	L	N
D	W	H	A	A	Q	I	N
E	F	G	K	V	E	P	R

2.2 BLOSUM Matrices

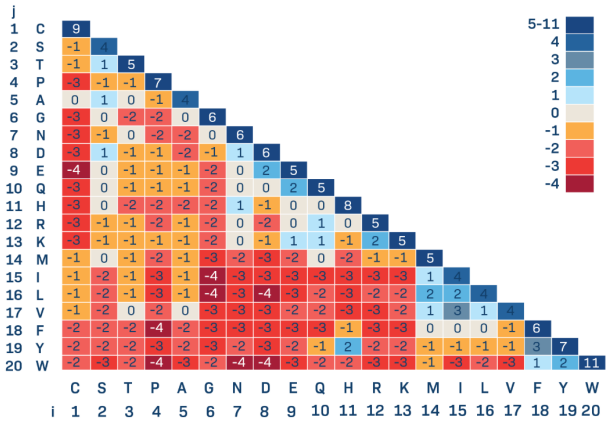
BLOSUM (BLOcks SUbstitution Matrix) refers to a family of matrices used to score similarity in peptide sequence alignments. These matrices assign scores to all possible substitutions of one amino acid for another, allowing the evaluation of the likelihood that two sequences are related.

The scores in BLOSUM matrices are numerical values known as *log-odds scores*. These scores represent a statistical measure of the probability that a pair of amino acids appears in an alignment due to a true biological relationship, compared to the probability that they appear together purely by chance.

Positive scores indicate that a pair of amino acids occurs more frequently than expected by chance, while negative scores indicate that the pair occurs less frequently than expected, suggesting that such a substitution is evolutionarily unfavorable or that the amino acids are highly dissimilar. A score of zero indicates that the pair appears exactly as frequently as would be expected at random [10].

The family of BLOSUM substitution matrices, particularly BLOSUM62, is the de facto standard in peptide database searches and sequence alignments [20].

Figure 3: Matriz BLOSUM 62



2.3 Distance Matrices

The distance matrix between backbone carbons is a two-dimensional representation that describes the spatial distances between pairs of residues in a protein, primarily using the Cβ atoms or, in the case of glycine, the Cα atoms. Each entry in the matrix corresponds to the Euclidean distance between the backbone carbons of two specific residues, allowing the three-dimensional organization of the polypeptide chain to be captured in a structured form. Due to the symmetric nature of distances between pairs of points, this matrix is symmetric and is useful for representing the packing and global geometry of proteins in structural prediction studies [9] [22] [23].

2.4 Contact Maps

Contact maps are two-dimensional representations that indicate the proximity between pairs of residues in the three-dimensional structure of a protein. They are modeled as a binary matrix where a value of 1 indicates that two residues are in contact (generally at a distance of less than 8 Å) and 0 indicates that they are not, allowing the capture of spatial interactions relevant to protein folding and stability [22].

2.5 ESM-2

ESM-2 is a family of large-scale language models developed by Meta AI, specifically designed to work with protein sequences. It is based on transformer architectures and is trained using masked language modeling, where the model learns to predict missing amino acids from the sequence context.

One of the emergent capabilities of ESM-2 is that, despite not being explicitly trained on three-dimensional structures, it learns internal representations that reflect structural information about proteins. In particular, its attention mechanisms capture relationships between pairs of amino acids that correlate with spatial contacts within the structure.

From these internal representations, it is possible to construct a probability matrix that reflects the likelihood of contact between pairs of residues. This matrix is derived from the model's attention maps, which are processed and transformed to estimate these probabilities, resulting in a representation that summarizes which regions of the protein are more likely to be close in three-dimensional space [15].

2.6 AlphaFold

AlphaFold is an artificial intelligence system developed by DeepMind that uses neural networks to predict the three-dimensional structure of a protein from its linear amino acid sequence. Its main objective is to translate this sequence into an accurate 3D structure, enabling an understanding of how a protein functions, how it interacts with other molecules, and how it might be modulated.

In CASP14 (Critical Assessment of Structure Prediction), AlphaFold achieved a historic performance by solving the protein folding problem with high accuracy. In this evaluation, it obtained results with an average deviation close to 0.96 Å, a level of accuracy comparable to experimental techniques such as X-ray crystallography.

The system is based on an architecture composed of two main components: the Evoformer, which processes evolutionary information and relationships between amino acids from related sequences, and the structure module, which generates the final 3D coordinates while respecting geometric and physical constraints [11].

2.7 Top L

The Top L is an evaluation metric in protein contact prediction that measures the precision of results by considering a number of residue pairs equal to the total length (L) of the protein sequence [21]. To calculate it, the following steps must be followed:

1. Assign a probability score to each predicted residue pair.
2. Sort all pairs in a descending list according to this score.
3. Select the top k contacts from the list, where k corresponds to the value of the chosen metric (for example, if it is Top L/5, the top L/5 contacts are selected).
4. Calculate the precision as the percentage of native (real) contacts present in that selection: Precision = $TP/(TP + FP)$.

This metric is primarily used in CASP experiments to rank the performance of participating groups [19].

2.8 Shannon Entropy

Shannon entropy measures the average amount of information generated by a random data source; higher entropy values indicate greater information content in new data [18]. For a set of events with probabilities $p_1, p_2, \ldots, p_n$, it is mathematically defined as:

$$H = -\sum p_i \log(p_i) \tag{1}$$

Shannon entropy is useful for comparing contact matrices because it allows quantifying how contacts are distributed along the protein, that is, whether they are concentrated in a few regions or distributed more uniformly [14].

2.9 AUC-PR

AUC-PR is a metric used to evaluate the performance of prediction methods such as binary classifiers, especially in imbalanced datasets, as is the case in protein contact prediction, where positive cases are scarce. This measure summarizes the trade-off between

precision and recall across different thresholds, providing a global indicator of model accuracy, where values close to 1 represent optimal performance [17].

2.10 dRMSD

Distance-based root mean square deviation (dRMSD) is a metric used to quantify the difference between a predicted protein structure and an experimental one based on their internal pairwise atomic distances [2]. It is calculated as

$$\text{dRMSD} = \sqrt{\frac{1}{N}\sum_{i<j}\left(d_{ij}^{(\text{p})} - d_{ij}^{(\text{e})}\right)^2} \tag{2}$$

where d_{ij} represents the distance between atoms i and j, and N is the total number of pairs considered. Unlike traditional RMSD, this metric does not require prior alignment between structures, as it is invariant to rotations, translations, and reflections. dRMSD evaluates differences across multiple length scales and strongly penalizes global deviations, making it useful for capturing both local and global structural discrepancies.

2.11 Pearson correlation

The comparison between matrices can be performed using Pearson or Spearman correlation, which allow quantifying the similarity between their elements under different assumptions about the relationship between variables [5]. Pearson correlation measures linear relationships using the original values, while Spearman correlation is a non-parametric measure that evaluates monotonic relationships based on data ranks, making it more robust to outliers and suitable when the relationship is not strictly linear. Pearson correlation is calculated as follows:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{3}$$

where x_i and y_i represent the individual observations of the variables X and Y, respectively; $\bar{x}$ and $\bar{y}$ are the means of those variables; and n is the total number of observations. The numerator corresponds to the covariance between X and Y, while the denominator is the product of their standard deviations, so the coefficient r measures the strength and direction of the linear relationship between both variables, taking values in the interval $[-1, 1]$.

2.12 GDT-TS

The RMSD (root mean square deviation) between corresponding $C\alpha$ atoms can provide a useful measure of global similarity between two slightly different conformations of the same protein. However, this metric can be misleading, as small local perturbations, such as changes in hinge-like regions between domains, can produce large increases in RMSD, giving the impression that the structures are globally very different even if they retain important local similarities. Therefore, it is necessary to complement this type of measure with approaches that consider similarity at a regional level.

In this context, GDT-TS (Global Distance Test Total Score) is a metric used to evaluate structural similarity between two proteins, typically between an experimental structure and a predicted one. It is based on the fraction of residues that can be superimposed within different distance thresholds, allowing for a more robust capture of both global and local structural similarity, without relying solely on a single global deviation measure such as RMSD.

It is calculated as the average of the percentages of residues that can be superimposed under four different distance thresholds. The formula is GDT-TS = $(P_1 + P_2 + P_4 + P_8)/4$ where P_d represents the percentage of residues in the reference structure that can be superimposed with the corresponding residues of the model under a distance threshold d (where d is 1, 2, 4, and 8 Ångströms).

GDT-TS has been extensively tested in CASP competitions, providing a reliable ranking for evaluating the quality of 3D predictions [24].

3 METHODOLOGY

The objective of this work is to generate N peptides whose sequences produce a contact map similar to that of the target peptide, with the purpose of evaluating whether guiding the search in this manner can lead to the identification of sequences with three-dimensional structures similar to that of the target peptide.

Algorithm 1 General Design Framework for Protein Sequence Optimization

Input: Set of target proteins $\mathcal{P}$, Prediction model $\mathcal{M}$
Output: 3D structural validation of top N generated sequences per target

```
for each target protein P_target ∈ 𝒫 do
    ▷ Phase 1: Data extraction and ground-truth construction
    FASTA ← RetrieveFASTA(P_target)
    Coord_3D ← ExtractCoordinates(P_target, C_β atoms)
    D_real ← ComputeDistanceMatrix(Coord_3D)
    L ← Length(FASTA)
    Initialize M_real as L × L zero matrix
    for i = 1 to L do
        for j = 1 to L do
            if D_real(i, j) < 8 Å then
                M_real(i, j) ← 1                ▷ Contact detected
            else
                M_real(i, j) ← 0
            end if
        end for
    end for
    ▷ Phase 2: Evolutionary Generation and Optimization
    Pop_init ← GenerateConstrainedPopulation(FASTA)
    Candidates ← GeneticAlgorithm(FASTA, M_real, Pop_init)
    ▷ Phase 3: Structural Validation via AlphaFold
    for each S_candidate ∈ Candidates do
        Structure_pred ← AlphaFold(S_candidate)
        Δ_3D ← ComputeRMSD(Structure_pred, Coord_3D)
        Record (S_candidate, Δ_3D)
    end for
end for
return Structural validation results
```

3.1 Genetic Algorithm

For the generation of the N peptides with a contact map similar to that of the target peptide, a genetic algorithm ($\mu + \lambda$) with a string-based encoding scheme was used.

$$\Sigma = G, A, V, L, I, M, P, F, W, S, T, C, Y, N, Q, D, E, K, R, H$$

where each character represents an amino acid according to the Table shown in Figure 2.

The peptides used were some of the was those used in [13], and they were obtained from [12] – KCM (GitHub).

1aay	1msl	1r5l	2prt	3h25
1emn	1n7d	1s7m	2qiw	3m9q
1f2i	1np6	2e45	2qq8	3sb1
1llm	1o6w	2i13	3c8v	3vdu
1lql	1p9n	2kxq	3clq	3w68
2lkm	2oar	3e7r	3ewk	4gzn
5uiy				

Table 2: Dataset of Target Peptides utilized in the experiments.

3.1.1 Initial Population. The initial population was generated pseudo-randomly, as the search space is very large and it is not worthwhile to start with an initial population that is not biologically meaningful or that has chemical properties very different from those of the target peptide.

For this purpose, two initial populations were used: one that respects the chemical properties shown in Figure 2 and avoids the changes listed in Table 1 (4), and another generated using the scores from the BLOSUM62 matrix, shown in Figure 3 (Algorithm 3).

The way in which the allowed mutations, used for the initial population defined by chemical properties, were specified is shown in the Table Algorithm 3.

Original AA	Permitted Substitutions
A, C, I, L, M, V	A, C, T, V, I, L, M, F
D, S	D, E, G, H, K, N, P, Q, R, S, Y
E, H, N, R	D, E, G, H, K, N, P, Q, R, S, W, Y
F	A, C, V, I, L, M, F
G	D, E, G, H, N, P, Q, R, S, W, Y
H	D, E, G, H, K, N, P, Q, R, S, W, Y
K, Q, Y	D, E, H, K, N, P, Q, R, S, W, Y
P	D, E, G, H, K, N, P, Q, S, W, Y
T	A, C, T, V, I, L, M
W	D, E, G, H, K, N, P, Q, R, W, Y

Table 3: Allowed Mutations for Initial Population Generation.

3.1.2 Parent Selection. For parent selection, tournament selection was used (5), with $k = 2$ in the initial generations and $k = 3$ in the later ones, with the aim of encouraging exploration at the beginning and promoting exploitation toward the end.

Algorithm 2 ESM-2 Based Genetic Algorithm

Input: Target protein P_{target}, Fitness function f, Generations G, Offspring count C, Population size P_{best}
Output: P_{best} unique sequences and evolutionary statistics

Initialize $Cache \leftarrow \emptyset$ ▷ Storage for memoization
$M_{target} \leftarrow$ LoadTargetContactMap(P_{target})
$S_{init} \leftarrow$ LoadInitialSequences(P_{target})
▷ *Initial Population Processing*
$Pop \leftarrow$ EvaluateESM2($S_{init}, M_{target}, f, Cache$)
Sort Pop by fitness descending and filter unique sequences
$P_0 \leftarrow Pop[1 : P_{best}]$
Update $Cache$ with fitness values from P_0
for $g \leftarrow 1$ to G **do**
▷ *Adaptive Tournament Size*
  if $g < \frac{2}{3}G$ **then**
    $k \leftarrow 2$
  else
    $k \leftarrow 3$
  end if
▷ *Selection and Variation*
  $Winners \leftarrow \{$Tournament(P_{g-1}, k) for $i = 1 \ldots C\}$
  $S_{offspring} \leftarrow \emptyset$
  for $i \leftarrow 1$ to C **do**
    $p_1, p_2 \leftarrow$ RandomSelect($Winners$)
    $child \leftarrow$ Crossover($p_1.seq, p_2.seq$)
    $mutated_child \leftarrow$ Mutation($child$)
    $S_{offspring} \leftarrow S_{offspring} \cup \{mutated_child\}$
  end for

  end for ▷ *Optimized Fitness Evaluation*
  $S_{new} \leftarrow \{s \in S_{offspring} \mid s \notin Cache\}$
  if $S_{new} \neq \emptyset$ **then**
    $Eval_{new} \leftarrow$ EvaluateESM2($S_{new}, M_{target}, f, Cache$)
    Update $Cache$ with $Eval_{new}$
  end if
  $P_{offspring} \leftarrow \{(s, Cache[s])$ for $s \in S_{offspring}\}$
▷ *Survival and Clone Filtering ($\mu + \lambda$)*
  $P_{combined} \leftarrow$ Sort($P_{g-1} \cup P_{offspring}$) by fitness descending
  $P_g \leftarrow \emptyset$, $Seen \leftarrow \emptyset$
  for each $ind \in P_{combined}$ **do**
    if $ind.seq \notin Seen$ **then**
      $P_g \leftarrow P_g \cup \{ind\}$
      $Seen \leftarrow Seen \cup \{ind.seq\}$
    end if
    if $|P_g| = P_{best}$ **then break**
    end if
  end for
  Compute and record statistics (Best, Mean, Median)
end for
Save results and execution time
return P_G

Algorithm 3 Initial Population Generation: BLOSUM62 Strategy

Input: Original sequence S, Population size N, BLOSUM62 matrix B
Output: Initial population P with N sequences

▷ *Evolutionary Sampling Loop*
for $i \leftarrow 1$ to N **do**
  Initialize empty sequence S_{new}
  for each amino acid c in S **do**
    ▷ *Probability distribution based on substitution scores*
    $Candidates \leftarrow \{j \mid B[c, j] > 0\}$
    $Weights \leftarrow \{B[c, j] + 1 \mid j \in Candidates\}$
    $Probs \leftarrow Weights / \sum Weights$
    $A_{new} \leftarrow \text{WeightedRandomChoice}(Candidates, Probs)$
    $S_{new} \leftarrow S_{new} \cup \{A_{new}\}$
  end for
  $P \leftarrow P \cup \{S_{new}\}$
end for
return P

Algorithm 4 Initial Population Generation: Hydrophobic Strategy

Input: Original sequence S, Population size N, Allowed mutations dictionary D
Output: Initial population P with N sequences

▷ *Property-Preserving Selection*
for $i \leftarrow 1$ to N **do**
  Initialize empty sequence S_{new}
  for each amino acid aa in S **do**
    ▷ *Random selection from compatible group*
    $mutated_aa \leftarrow \text{RandomChoice}(D[aa])$
    $S_{new} \leftarrow S_{new} \cup \{mutated_aa\}$
  end for
  $P \leftarrow P \cup \{S_{new}\}$
end for
return P

Algorithm 5 Tournament Selection Operator

Input: Population P, Tournament size k
Output: Selected individual i^*

$C \leftarrow \{c_1, c_2, \ldots, c_k\} \subseteq P$ ▷ Sample k individuals
$i^* \leftarrow \{c \in C \mid f(c) = \max_{x \in C} f(x)\}$ ▷ Select the individual with the highest fitness
return i^*

3.1.3 Genetic Crossover. The classical two-point crossover was used (6), generating one offspring per crossover.

3.1.4 Genetic Mutation. A uniform mutation with a probability of 0.05 was used (7), with the constraint that polar amino acids could only mutate into polar amino acids and vice versa, according to Figure 2.

3.1.5 Fitness. To calculate the fitness, it was necessary to find a methodology that would allow the comparison of the contact matrices or maps of the target peptides with the contact probability matrices provided by the ESM-2 model. For this purpose, two fitness measures were used. The first step was to filter out the obvious contacts. In protein structure, amino acids that are adjacent in the chain (for example, the amino acid at position i and its immediate neighbors at $i+1$ or $i+2$) will always be in physical contact because they are chemically bonded. Predicting this is trivial and does not provide useful information about the real 3D folding of the protein. The purpose of filtering these contacts was to ignore these evident local contacts in order to force the genetic algorithm to focus on predicting long-range contacts, which are the ones that truly define the complex three-dimensional shape of the protein. A separation distance of 3 was used. This means that if the distance between two amino acids is less than 3 positions in the linear chain, their contact probability is set to 0. Once the obvious contacts were masked, one of the following fitness measures was used. In this step, the lower triangular portion of the matrix is also set to zero. Since it is a symmetric contact probability matrix, this part contains only redundant information.

Algorithm 6 Two-Point Crossover

Input: Parents S_1, S_2 of length n
Output: Offspring S_{child}

$p_1 \leftarrow \text{UniformRandom}(0, n-2)$ ▷ Select the first crossover point
$p_2 \leftarrow \text{UniformRandom}(p_1 + 1, n)$ ▷ Select the second crossover point
$r \leftarrow \text{UniformRandom}(0, 1)$ ▷ Random value for parent role assignment
if $r < 0.5$ **then**
  $(A, B) \leftarrow (S_1, S_2)$
else
  $(A, B) \leftarrow (S_2, S_1)$
end if
$S_{child} \leftarrow A[0 \ldots p_1 - 1] \parallel B[p_1 \ldots p_2 - 1] \parallel A[p_2 \ldots n - 1]$ ▷ Resulting crossover
return S_{child}

Algorithm 7 Mutation

Input: Sequence S of length n, Mutation probability p_m
Input: Sets of amino acids $\mathcal{A}_{pol}$ (polar) and $\mathcal{A}_{non}$ (non-polar)
Output: Mutated sequence S'

$S' \leftarrow S$
for $i \leftarrow 0$ **to** $n - 1$ **do**
  $r \leftarrow \text{UniformRandom}(0, 1)$
  if $r < p_m$ **then** ▷ Check if mutation occurs at this locus
    if $S[i] \in \mathcal{A}_{pol}$ **then**
      $S'[i] \leftarrow \text{RandomChoice}(\mathcal{A}_{pol} \setminus \{S[i]\})$ ▷ Replace with a different polar AA $S[i] \in \mathcal{A}_{non}$
      $S'[i] \leftarrow \text{RandomChoice}(\mathcal{A}_{non} \setminus \{S[i]\})$ ▷ Replace with a different non-polar AA
    end if
  end if
end for
return S'

Fitness Shannon Entropy

It evaluates how well the correct predictions (true contacts correctly predicted) are distributed throughout the entire protein structure. Instead of only counting how many correct predictions there are, it uses Shannon Entropy to reward models that are able to predict contacts across diverse regions of the protein, avoiding the concentration of all correct predictions within a single block.

If the value of H is high means that the correctly predicted contacts are well distributed across the entire 5×5 grid. This is a good indicator that the model captures the global structure of the protein. And if the value of H is low means that, even if there are correct predictions, they are concentrated in only a few regions of the grid.

Given how the grid is defined, the best possible value that can be obtained is $ln(25) \approx 3.2188$. To reach the absolute maximum value in this function, two conditions must be satisfied simultaneously: 100% correct predictions and a perfectly uniform distribution of those predictions. The complete algorithm is shown in 8.

Algorithm 8 Fitness Calculation: Shannon Entropy

Input: Predicted probability matrix M_{pred}, real contact matrix M_{real}, protein length L, grid size G (default 5)
Output: Shannon entropy fitness score H

```
▷ 1. Selection of Top L Predictions
FlatPred ← Flatten(M_pred)
TopL ← Indices of L largest values in FlatPred
▷ 2. Identification of True Positives (TP)
TP_Coords ← ∅
for each index (r, c) ∈ TopL do
    if M_real[r, c] = 1 then
        TP_Coords ← TP_Coords ∪ {(r, c)}
    end if
end for
▷ 3. Proportional Mapping to G × G Grid
Initialize G × G grid T with zeros
for each (r, c) ∈ TP_Coords do
    i ← ⌊(r · G)/L⌋
    j ← ⌊(c · G)/L⌋
    T[i, j] ← T[i, j] + 1
end for
▷ 4. Shannon Entropy Computation
H ← 0
for each count ∈ Flatten(T) do
    if count > 0 then
        p_e ← count/L
        H ← H − (p_e · ln(p_e))
    end if
end for
return H
```

Fitness AUC PR

This metric operates through a dynamic process that ranks the model's predictions from highest to lowest probability in order to evaluate the quality of the ranking using a moving threshold. Its logic consists of traversing this list while computing precision and recall at each step, capturing the model's ability to concentrate true contacts at the top positions and its robustness against false positives. Mathematically, it represents the area under the precision–recall curve through a weighted sum, where the best possible value is 1.0. Achieving this maximum score means that the model attains a perfect ranking, successfully identifying all true contacts in the protein before making the first error, thereby ensuring a complete separation between biological signal and statistical noise. The complete algorithm is shown in 9.

Algorithm 9 Fitness Calculation: AUC-PR

Input: Predicted probability matrix M_{pred}, real contact matrix M_{real}
Output: AUC-PR fitness score A

```
▷ 1. Vectorization and Alignment
P ← [M_pred[i, j] for (i, j) ∈ M_∇]↓↕]      ▷ Predicted scores
R ← [M_real[i, j] for (i, j) ∈ M_∇]↓↕]      ▷ Binary ground truth
▷ 2. Confidence Ranking
Sort pairs in (P, R) in descending order based on the values in P
▷ 3. Precision-Recall Curve Integration
TP ← 0                                      ▷ True Positives counter
TotalReal ← ∑ R                             ▷ Total number of real contacts
A ← 0
Recall_prev ← 0
for n ← 1 to |P| do
    if R_n = 1 then
        TP ← TP + 1
    end if
    Precision_n ← TP/n
    Recall_n ← TP/TotalReal
    ▷ Update Area (Average Precision traversal)
    A ← A + (Recall_n − Recall_prev) × Precision_n
    Recall_prev ← Recall_n
end for
return A
```

3.2 Models

Since there are two ways to generate the initial population and two ways to evaluate the fitness, there are four models; *blosum62-fitness-AUC-PR* (model 1), *blosum62-fitness-shannon-entropy* (model 2), *hydrophobic-fitness-AUC-PR* (model 3) and *hydrophobic-fitness-shannon-entropy* (model 4). However, since obtaining the contact probability matrix for a given sequence takes approximately 0.1 seconds, it was decided to use only two models.

3.2.1 Model selection. For model selection, 3 out of the 31 peptides were randomly selected, resulting in 1llm, 1s7m, and 3sb1. For each case, each model was run with an initial population of 50 individuals, 30 generations, and 30 offspring, maintaining a population of 25 individuals. The initial populations were generated using

the same seed; thus, the *blosum62-fitness-AUC-PR* and *blosum62-fitness-shannon-entropy* models started with the same initial population, and the *hydrophobic-fitness-AUC-PR* and *hydrophobic-fitness-shannon-entropy* models also started with the same initial population. For each model, the best sequence from the final population (according to its respective fitness) was selected, and its 3D structure was obtained using the AlphaFold2 predictor. Finally, the predicted distance matrix was compared with the real distance matrix. Table 4 shows the obtained results.

Peptide	Model	dRMSD	Correlation
1	1s7m	2.4217	0.9263
	3sb1	0.5291	0.9961
	1llm	1.6642	0.9475
	Average	**1.5383**	**0.9566**
2	1s7m	2.6752	0.9116
	3sb1	1.9875	0.9730
	1llm	1.9519	0.9254
	Average	**2.2049**	**0.9367**
3	1s7m	1.4607	0.9741
	3sb1	1.3961	0.9780
	1llm	4.2547	0.6455
	Average	**2.3705**	**0.8659**
4	1s7m	3.3262	0.8930
	3sb1	1.4622	0.9754
	1llm	4.8886	0.6686
	Average	**3.2257**	**0.8457**

Table 4: Performance results using one candidate sequence per model and peptide

The two models with the lowest dRMSD and highest correlation were Models 1 and 2. However, the dRMSD values for Models 2 and 3 were very similar, which motivated a secondary analysis to determine which two models should be retained among these three candidates. For this analysis, the final population—ordered by fitness—was evaluated at representative ranks corresponding to individuals 1 (best), 2, 3, 12 (median), and 25 (worst), with the results summarized in Table 5. Overall, Models 1 and 2 consistently showed the most favorable performance across these evaluations.

It is important to note that both models were derived from the same initial population (based on the BLOSUM62 substitution matrix), differing only in the fitness evaluation strategy used during optimization.

The final models (*blosum62-fitness-AUC-PR* and *blosum62-fitness-shannon-entropy*) were run on all 31 peptides (Table 2), but now using an initial population of 200 individuals, 200 generations, and 120 offspring, while maintaining a population of the 100 best individuals per generation.

For each model and each peptide, individuals 1, 15, 30, 45, 60, 75 and 90 from the final population were selected, ordered from best to worst fitness. For each of them, a 3D structure prediction was obtained

Model	Peptide	dRMSD (Avg)	Correlation (Avg)
1	1s7m	2.9918	0.9144
	3sb1	0.6883	0.9936
	1llm	1.7068	0.9463
	Average	**1.7957**	**0.9517**
2	1s7m	2.6182	0.9146
	3sb1	1.7814	0.9751
	1llm	1.9222	0.9316
	Average	**2.1073**	**0.9404**
3	1s7m	2.2284	0.9386
	3sb1	1.3860	0.9777
	1llm	3.5153	0.7477
	Average	**2.3766**	**0.8880**

Table 5: Performance results using 5 candidate sequences per model and peptide

using AlphaFold2, and from this, the corresponding distance matrix was derived. Three evaluation metrics were computed: dRMSD, Pearson correlation, and GDT-TS. It is important to note that dRMSD and Pearson correlation were calculated by comparing the distance matrix derived from the predicted structure against the distance matrix of the target peptide sequence, whereas GDT-TS was computed directly from the predicted three-dimensional coordinates

Additionally, to evaluate the performance of the models, seven random sequences were generated for each target peptide (with the same number of amino acids as the target peptide), and the previously mentioned metrics were computed.

3.3 Execution environment

All experiments were implemented in Python and executed on a single GPU. The hardware configuration used for all computational runs was:

- **CPU:** Intel(R) Core(TM) i5-9400 CPU at 2.90GHz.
- **RAM:** 32GB.
- **GPU:** NVIDIA GeForce RTX 3090 Ti.
- **OS:** Pop!_OS 22.04 LTS.

The Genetic Algorithm used full-batch processing throughout; no data loading parallelism was employed to ensure search trajectory consistency. Reported runtimes are wall-clock times averaged over the experimental runs.

4 EXECUTION TIME

The execution of the genetic algorithms took, on average, 40.5 minutes per peptide, corresponding to an average of 20.9 hours per model.

The average execution time for AlphaFold2 to predict the seven sequences per protein was 418.18 seconds, corresponding to approximately 3.6 hours per model. Considering the three models (Shannon entropy, AUC-PR, and random), this results in a total computational time of approximately 10.8 hours.

Therefore, approximately 53 hours are required to replicate the experiments.

5 RESULTS

First, it was observed that some sequences with lower fitness exhibited better results when comparing their predicted structures than those with higher fitness, which suggests that contact maps alone may not fully capture all aspects of three-dimensional structural similarity.

For this reason, two complementary evaluation perspectives were adopted in order to address different but equally relevant questions about model performance. On the one hand, a *best-case performance approach* was considered, in which, for each model, the sequence achieving the best structural similarity (measured by dRMSD or GDT-TS) among the sampled candidates was selected. This perspective evaluates the exploratory capability of each method, that is, its ability to generate high-quality structural solutions regardless of how they are ranked by the fitness function.

On the other hand, a *robustness-based approach* was employed by summarizing the structural performance of the sampled population using the median dRMSD and GDT-TS values. This allows assessing the overall quality of the solutions produced by each model and provides insight into whether the search process consistently converges toward regions of the sequence space associated with structurally accurate conformations, rather than relying on isolated high-performing individuals.

Together, these two perspectives provide a more comprehensive evaluation, capturing both the best-case performance and the general behavior of each model.

Given that both models share the same initial population and differ only in the fitness evaluation strategy, this analysis reduces to addressing the following question: which of the two fitness evaluation schemes yields better performance?

5.1 Best-case performance

To determine whether there was a statistically significant difference among the three models, the individual with the highest GDT-TS value was selected for each peptide (figure 5) and the Friedman test was performed. The same procedure was applied to dRMSD (figure 4).

The results of the Friedman test indicate statistically significant differences among the three evaluated models for both dRMSD (statistic = 7.806, p = 0.0201) and GDT-TS (statistic = 14.9391, p = 0.00057). Based on these results, pairwise comparisons using the Wilcoxon test were conducted.

Under the post-hoc Wilcoxon analysis for GDT-TS, no statistically significant differences were observed between the two fitness evaluation strategies (p = 0.24882). In contrast, both approaches showed significant improvements over the random baseline, with p-values of 0.00036 (AUC-PR) and 0.00159 (Shannon entropy), respectively.

For dRMSD, no significant differences were detected between the two strategies (p = 0.14060), nor between the AUC-PR-based approach and the random baseline (p = 0.06642). However, the Shannon entropy–based strategy did show a statistically significant difference compared to the random baseline (p = 0.000541).

Figure 4: dRMSD by model for the individuals with the lowest dRMSD value in each peptide.

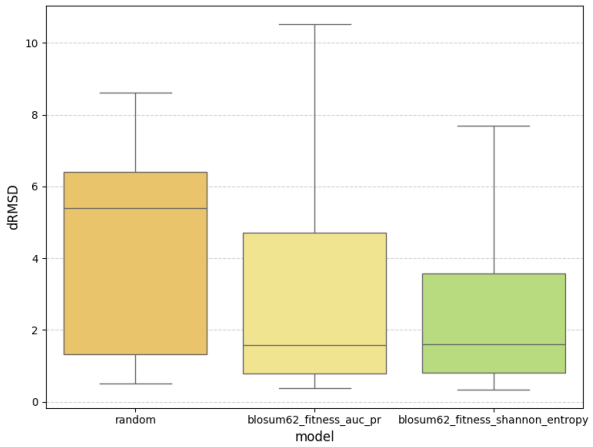
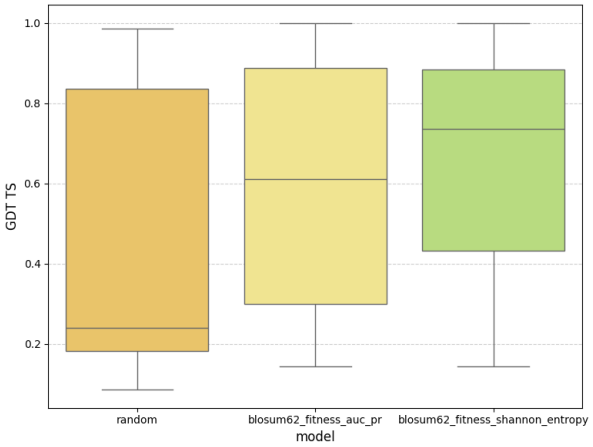


Figure 5: GDT-TS by model for the individuals with the highest GDT-TS value in each peptide.



Overall, these results indicate that while both strategies behave similarly relative to each other, only the Shannon entropy–based formulation shows a consistent statistically significant improvement over the random baseline across both evaluation metrics.

5.2 Best-robustness performance

To evaluate the overall quality and consistency of the populations generated by the models, the median GDT-TS value of the sampled individuals was calculated for each peptide and the Friedman test was performed. The same procedure was applied to the median dRMSD.

The results of the Friedman test indicate statistically significant differences among the three evaluated models for both the median dRMSD (statistic = 17.871, p = 0.00013) and the median GDT-TS (statistic = 26.298, $p = 1.948 \times 10^{-6}$). Based on these results, pairwise comparisons using the Wilcoxon test were conducted.

Figure 6: Median dRMSD by model for each peptide population.

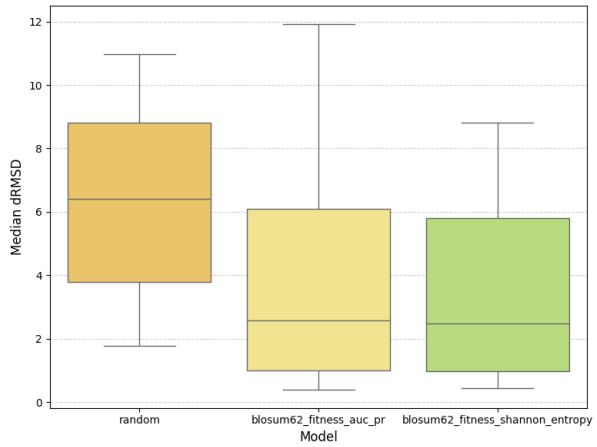
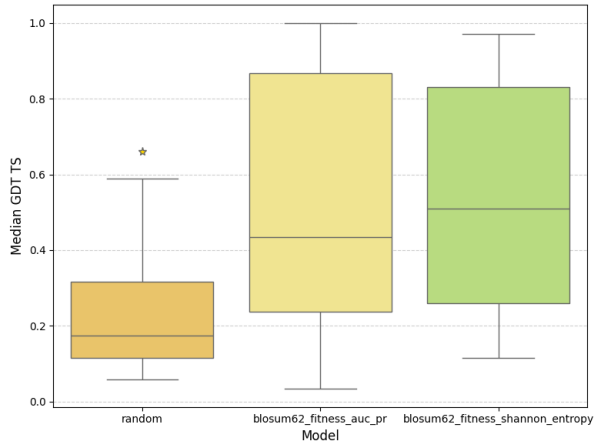


Figure 7: Median GDT-TS by model for each peptide population.



Under the post-hoc Wilcoxon analysis for the median GDT-TS, no statistically significant differences were observed between the two fitness evaluation strategies ($p = 0.75767$). In contrast, both approaches showed highly significant improvements over the random baseline, with p-values of 4.163×10^{-7} (AUC-PR) and 1.080×10^{-5} (Shannon entropy), respectively.

Similarly, for the median dRMSD, no significant differences were detected between the two fitness evaluation strategies ($p = 0.51676$). However, unlike the best-case performance scenario, both the AUC-PR-based approach ($p = 0.000145$) and the Shannon entropy-based strategy ($p = 1.2 \times 10^{-5}$) demonstrated a statistically significant improvement compared to the random baseline.

Overall, these results indicate that, from a population robustness perspective, both fitness evaluation strategies perform similarly to each other and consistently outperform the random baseline across both structural evaluation metrics. This suggests that both models effectively guide the search towards regions of the sequence space with higher structural quality, rather than merely producing individuals with random sequences

5.3 Statistical summary

Following the inferential analyses conducted in the previous sections, a global characterization of the structural performance of the evaluated models was performed. While statistical tests provide evidence regarding differences between models, they do not fully describe the magnitude and variability of their performance across peptides and evaluation metrics. For this reason, a descriptive statistical summary was computed for each model, considering GDT-TS, dRMSD, and Pearson correlation across all evaluated peptides. This allows a more comprehensive assessment of the overall behavior of each approach in terms of structural quality and prediction consistency.

These results are presented in Tables 6, 7 and 8.

Table 6: Statistical summary of GDT-TS (Mean ± SD) per peptide for the two fitness evaluation strategies.

Peptide	Model with AUC-PR	Model with Shannon Entropy
1aay	0.093 ± 0.028	0.297 ± 0.302
1emn	0.528 ± 0.083	0.438 ± 0.174
1f2i	0.198 ± 0.066	0.246 ± 0.111
1llm	0.153 ± 0.018	0.643 ± 0.060
1lql	0.740 ± 0.054	0.791 ± 0.053
1msl	0.542 ± 0.018	0.401 ± 0.105
1n7d	0.233 ± 0.055	0.246 ± 0.057
1np6	0.279 ± 0.014	0.290 ± 0.080
1o6w	0.129 ± 0.025	0.128 ± 0.021
1p9n	0.091 ± 0.082	0.456 ± 0.297
1r5l	0.343 ± 0.036	0.217 ± 0.032
1s7m	0.408 ± 0.049	0.504 ± 0.013
2e45	0.432 ± 0.224	0.630 ± 0.346
2i13	0.131 ± 0.016	0.607 ± 0.262
2kxq	0.608 ± 0.199	0.493 ± 0.176
2lkm	0.840 ± 0.035	0.855 ± 0.016
2oar	0.992 ± 0.012	0.977 ± 0.017
2prt	0.867 ± 0.033	0.190 ± 0.106
2qiw	0.870 ± 0.018	0.776 ± 0.039
2qq8	0.423 ± 0.036	0.406 ± 0.043
3c8v	0.889 ± 0.017	0.750 ± 0.032
3clq	0.750 ± 0.043	0.930 ± 0.028
3e7r	0.340 ± 0.107	0.226 ± 0.083
3ewk	0.981 ± 0.007	0.935 ± 0.069
3h25	0.868 ± 0.028	0.914 ± 0.015
3m9q	0.910 ± 0.079	0.870 ± 0.158
3sb1	0.932 ± 0.007	0.852 ± 0.051
3vdu	0.140 ± 0.026	0.197 ± 0.023
3w68	0.331 ± 0.068	0.333 ± 0.027
4gzn	0.273 ± 0.235	0.332 ± 0.377
5uiy	0.711 ± 0.011	0.724 ± 0.011

Additionally, to complement the population-level summary, an analysis of the best-performing individual for each peptide and model was conducted based on the GDT-TS metric. For each case, the sequence achieving the highest GDT-TS value was selected, allowing a focused evaluation of the most favorable structural outcome produced by each approach. This provides further insight into

Table 7: Statistical summary of dRMSD (Mean ± SD) per peptide for the two fitness evaluation strategies.

Peptide	Model with AUC-PR	Model with Shannon Entropy
1aay	12.163 ± 1.020	7.997 ± 5.004
1emn	2.270 ± 0.991	4.689 ± 2.450
1f2i	7.455 ± 2.646	5.779 ± 1.232
1llm	7.172 ± 0.280	1.951 ± 0.158
1lql	1.217 ± 0.183	1.094 ± 0.432
1msl	1.909 ± 0.131	3.138 ± 1.417
1n7d	5.811 ± 1.101	5.453 ± 0.790
1np6	5.919 ± 0.577	5.642 ± 1.013
1o6w	10.081 ± 0.622	8.974 ± 1.492
1p9n	10.019 ± 4.721	6.458 ± 7.919
1r5l	5.087 ± 0.619	7.054 ± 0.881
1s7m	2.556 ± 0.257	2.505 ± 0.047
2e45	4.083 ± 2.636	3.960 ± 4.032
2i13	7.689 ± 0.139	3.110 ± 2.602
2kxq	2.268 ± 1.368	3.831 ± 1.682
2lkm	0.924 ± 0.242	0.793 ± 0.087
2oar	0.430 ± 0.057	0.424 ± 0.063
2prt	0.747 ± 0.039	7.274 ± 1.176
2qiw	0.873 ± 0.077	1.044 ± 0.088
2qq8	4.308 ± 0.468	5.824 ± 0.443
3c8v	0.814 ± 0.028	1.194 ± 0.176
3clq	1.393 ± 0.092	0.704 ± 0.153
3e7r	3.874 ± 0.408	6.110 ± 2.853
3ewk	0.515 ± 0.095	0.770 ± 0.334
3h25	1.095 ± 0.155	0.934 ± 0.095
3m9q	0.892 ± 0.246	1.244 ± 1.260
3sb1	0.559 ± 0.030	1.209 ± 0.442
3vdu	7.651 ± 0.532	5.969 ± 1.444
3w68	4.838 ± 1.155	3.169 ± 0.843
4gzn	7.547 ± 2.945	6.439 ± 3.822
5uiy	1.814 ± 0.138	1.662 ± 0.046

Table 8: Statistical summary of Pearson Correlation (Mean ± SD) per peptide for the two fitness evaluation strategies.

Peptide	Model with AUC-PR	Model with Shannon Entropy
1aay	0.453 ± 0.014	0.644 ± 0.179
1emn	0.914 ± 0.106	0.621 ± 0.279
1f2i	0.569 ± 0.092	0.577 ± 0.079
1llm	0.656 ± 0.016	0.931 ± 0.013
1lql	0.979 ± 0.007	0.981 ± 0.018
1msl	0.960 ± 0.004	0.831 ± 0.204
1n7d	0.758 ± 0.088	0.796 ± 0.076
1np6	0.759 ± 0.026	0.769 ± 0.046
1o6w	0.551 ± 0.015	0.315 ± 0.042
1p9n	0.507 ± 0.169	0.785 ± 0.278
1r5l	0.872 ± 0.010	0.866 ± 0.005
1s7m	0.918 ± 0.015	0.921 ± 0.003
2e45	0.771 ± 0.173	0.796 ± 0.241
2i13	0.447 ± 0.002	0.812 ± 0.154
2kxq	0.881 ± 0.117	0.740 ± 0.153
2lkm	0.983 ± 0.009	0.988 ± 0.003
2oar	0.998 ± 0.001	0.997 ± 0.001
2prt	0.986 ± 0.001	0.571 ± 0.025
2qiw	0.990 ± 0.002	0.984 ± 0.003
2qq8	0.781 ± 0.026	0.819 ± 0.011
3c8v	0.992 ± 0.001	0.987 ± 0.002
3clq	0.982 ± 0.003	0.996 ± 0.001
3e7r	0.639 ± 0.096	0.443 ± 0.207
3ewk	0.995 ± 0.001	0.992 ± 0.006
3h25	0.986 ± 0.003	0.990 ± 0.002
3m9q	0.990 ± 0.006	0.986 ± 0.016
3sb1	0.996 ± 0.000	0.980 ± 0.012
3vdu	0.330 ± 0.028	0.414 ± 0.115
3w68	0.804 ± 0.157	0.857 ± 0.011
4gzn	0.718 ± 0.118	0.700 ± 0.193
5uiy	0.967 ± 0.001	0.968 ± 0.001

the upper-bound capability of each model in terms of structural quality. The results are presented in Table 13.

When analyzing the best-performing individuals per peptide according to GDT-TS using a threshold of GDT-TS> 0.9, the AUC-PR-based model identifies 6 peptides, while the Shannon entropy-based model identifies 8. Both models coincide in 4 peptides: 2oar, 3ewk, 3m9q, and 3sb1. The remaining peptides are uniquely identified by each model: AUC-PR additionally captures 2prt and 3c8v, whereas Shannon identifies 3clq, 3h25, 2e45, and 4gzn.

The information for the sequences with GDT-TS> 0.9 in which both models coincided is presented in Table 10.

Additionally, an analogous analysis was performed for sequences with low structural similarity, considering a threshold of GDT-TS< 0.1, in order to identify the peptides for which each model produced the poorest structural predictions. The AUC-PR-based model produced 14 such cases, while the Shannon entropy-based model produced 10. Both approaches coincided in 9 peptides (1o6w, 1f2i, 3w68, 2qq8, 1np6, 3vdu, 1r5l, 3e7r, and 1n7d). The results for the peptides in which both models failed are presented in Table 11.

5.4 Relationship between fitness and structural quality

To further assess the relationship between the optimization objective and the resulting structural quality, a Spearman correlation analysis was performed to quantify the association between fitness values and structural evaluation metrics. Specifically, correlations between fitness and GDT-TS, dRMSD, and Pearson correlation were computed for all evaluated individuals across both models. This allowed us to assess whether higher fitness values are consistently associated with improved structural quality, independently of linearity assumptions, and to quantify the extent to which the fitness function reflects true structural similarity in 3D space as well as its consistency across different structural descriptors. The results are presented in Table 12.

The results indicate a weak and inconsistent relationship between fitness and structural performance. For both models, correlations with GDT-TS and Pearson correlation are low, suggesting that the fitness function does not reliably capture global structural similarity. In the case of dRMSD, a moderate negative correlation is observed, indicating that higher fitness is only partially associated with improved structural agreement. Overall, these findings reinforce that the fitness function provides a limited proxy for true

Table 9: Comparison of peak structural performance (Maximum GDT-TS) and the population index (ind) for each peptide and model.

Peptide	AUC-PR: GDT-TS (ind)	Shannon: GDT-TS (ind)
1aay	0.1442 (1)	0.7404 (30)
1emn	0.6111 (1)	0.6574 (15)
1f2i	0.2500 (30)	0.4286 (75)
1llm	0.1897 (75)	0.7155 (15)
1lql	0.8077 (30)	0.8269 (75)
1msl	0.5694 (30)	0.5000 (1)
1n7d	0.2946 (1)	0.3304 (15)
1np6	0.3056 (30)	0.4352 (75)
1o6w	0.1635 (90)	0.1442 (1)
1p9n	0.2328 (90)	0.6466 (1)
1r5l	0.3833 (90)	0.2667 (15)
1s7m	0.4911 (60)	0.5179 (1)
2e45	0.8804 (1)	0.9239 (60)
2i13	0.1562 (15)	0.8229 (1)
2kxq	0.8417 (15)	0.7500 (1)
2lkm	0.8676 (45)	0.8676 (15)
2oar	1.0000 (1)	1.0000 (15)
2prt	0.9196 (45)	0.3125 (1)
2qiw	0.8947 (90)	0.8158 (1)
2qq8	0.4737 (60)	0.4737 (15)
3c8v	0.9079 (30)	0.8158 (60)
3clq	0.8295 (30)	0.9659 (15)
3e7r	0.4833 (1)	0.2833 (45)
3ewk	0.9853 (30)	0.9853 (15)
3h25	0.9000 (30)	0.9250 (15)
3m9q	0.9559 (15)	0.9706 (1)
3sb1	0.9375 (1)	0.9250 (90)
3vdu	0.1957 (90)	0.2283 (30)
3w68	0.4500 (45)	0.3667 (90)
4gzn	0.8036 (60)	0.9018 (90)
5uiy	0.7237 (1)	0.7368 (30)

Table 10: Sequence and structural quality (GDT-TS) comparison for high-performing overlapping peptides.

Peptide	Model	GDT-TS	Sequence
2oar	AUC-PR	1.0000	HTIFIIYEIREMWRERR
2oar	Shannon Entropy	1.0000	TYVFIVREIQKWLKSRK
3ewk	AUC-PR	0.9853	REAEEHIWKMNEWMDKH
3ewk	Shannon Entropy	0.9853	KRAEAKAWKAQEAMDGQ
3m9q	AUC-PR	0.9559	NNQQYQMYQEAMQAMQA
3m9q	Shannon Entropy	0.9706	SKQQQEMEEKWAKAAKG
3sb1	AUC-PR	0.9375	MPKWTWCAMYVVLIVQMAQM
3sb1	Shannon Entropy	0.9250	MGEIDWAMGQALMAIQMAQM

structural quality, with its predictive power varying depending on the chosen metric and model formulation.

To further evaluate the reliability of the fitness function as a final selection criterion, a top-1 coincidence analysis was performed to assess how often the fitness-selected individual corresponds to the best structural solution in terms of GDT-TS and dRMSD. The results show consistently low agreement, with coincidence rates ranging from 25.8% to 29.0% for GDT-TS and from 16.1% to 19.3% for dRMSD across both models, indicating that in approximately 70–80% of cases the fitness-based top-ranked solution does not match the true structural optimum. This suggests that the fitness function is more effective as a guide toward promising regions of the sequence space than as a definitive selector of the best structure. The results are presented in Table 13.

Table 11: Sequence and structural quality comparison for peptides where the best individuals failed to reach the 0.5 GDT-TS threshold.

Peptide	Model	GDT-TS	Sequence
1f2i	AUC-PR	0.2500	KPYDCCFTQCRTFDKKDHLKRSFDIREM
1f2i	Shannon Entropy	0.4286	RGCKCAGRDCKRFKRTDQLTKHFNVHKG
1n7d	AUC-PR	0.2946	DEGVCNKTDKCMDGNDYKSCVVFTFWLN
1n7d	Shannon Entropy	0.3304	STFICNSYDECADGSQECNVVVFTFMLY
1np6	AUC-PR	0.3056	ELRKAGAVKTIIYNDDHWWLWEDNPNE
1np6	Shannon Entropy	0.4352	YLKKAGATEYIIETDERWFIFSNYPND
1o6w	AUC-PR	0.1635	WKDGYDWNGQFYYVRDLSSKPSWNDP
1o6w	Shannon Entropy	0.1442	WSYAYDHPGEGWYYNHVTQKNTEHEP
1r5l	AUC-PR	0.3833	GQAEPQMDMDGMWGM
1r5l	Shannon Entropy	0.2667	CQPKPKADFSFGWLF
2qq8	AUC-PR	0.4737	YGNMDPMMYMHQYAAAHKR
2qq8	Shannon Entropy	0.4737	YGYMNPMMDAKYYFPPQFQ
3e7r	AUC-PR	0.4833	TEAECQKNCYAIKGAKGGDCVYPGEACDIR
3e7r	Shannon Entropy	0.2833	SQAHCETHCRIMHGTKGGKWAKGGAVCQCG
3vdu	AUC-PR	0.1957	GYLCYGCFEVKEEDWCYPCWEEE
3vdu	Shannon Entropy	0.2283	IDECKLCHHITNQDVCKPCWEKE
3w68	AUC-PR	0.4500	FHPPPQPDMDGMIGM
3w68	Shannon Entropy	0.3667	MDICFYLYWKKWMCL

Table 12: Spearman Rank Correlation Analysis between Fitness and Structural Metrics for GA Models.

Model	Metric	Spearman ρ	p-value
AUC-PR	GDT-TS	0.5552	6.08×10^{-19}
AUC-PR	dRMSD	-0.5199	2.01×10^{-16}
AUC-PR	Pearson Corr	0.5549	6.34×10^{-19}
Shannon Entropy	GDT-TS	0.0345	0.613
Shannon Entropy	dRMSD	-0.0153	0.822
Shannon Entropy	Pearson Corr	-0.0504	0.460

Table 13: Frequency of Coincidence between Top-1 Fitness and Top-1 Structural Metrics (%).

Model	Coincidence GDT-TS (%)	Coincidence dRMSD (%)
AUC-PR	25.81	19.35
Shannon Entrophy	29.03	16.13

6 CONCLUSIONS AND REFLECTIONS

6.1 General Conclusions

The results show that heuristic optimization approaches based on contact map surrogates are effective in guiding the search toward regions of the sequence space associated with improved structural quality. Both the AUC-PR and Shannon entropy-based strategies demonstrated statistically significant improvements over a random baseline in terms of population robustness, as assessed through median GDT-TS and dRMSD values.

Figure 8: Relationship between fitness and dRMSD across evaluated individuals for both models.

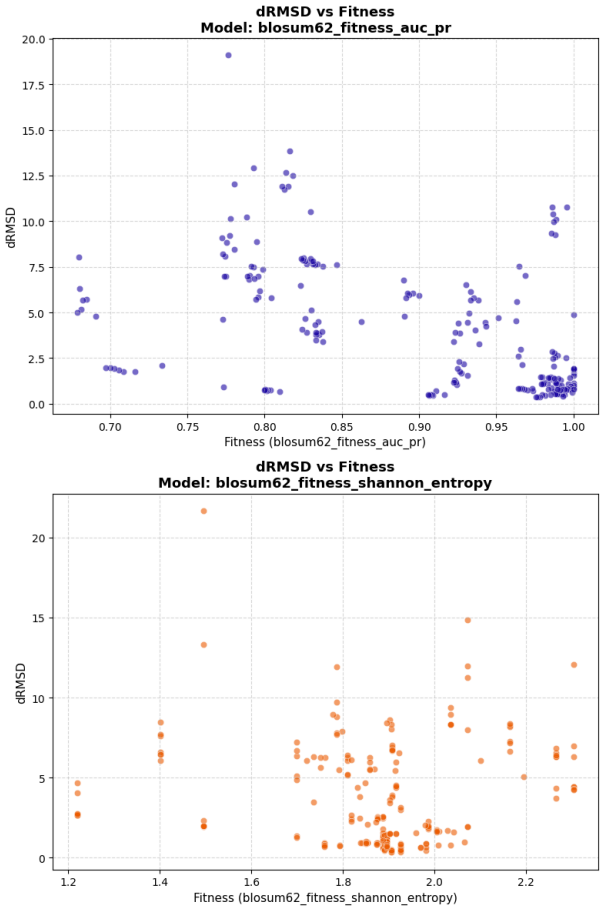
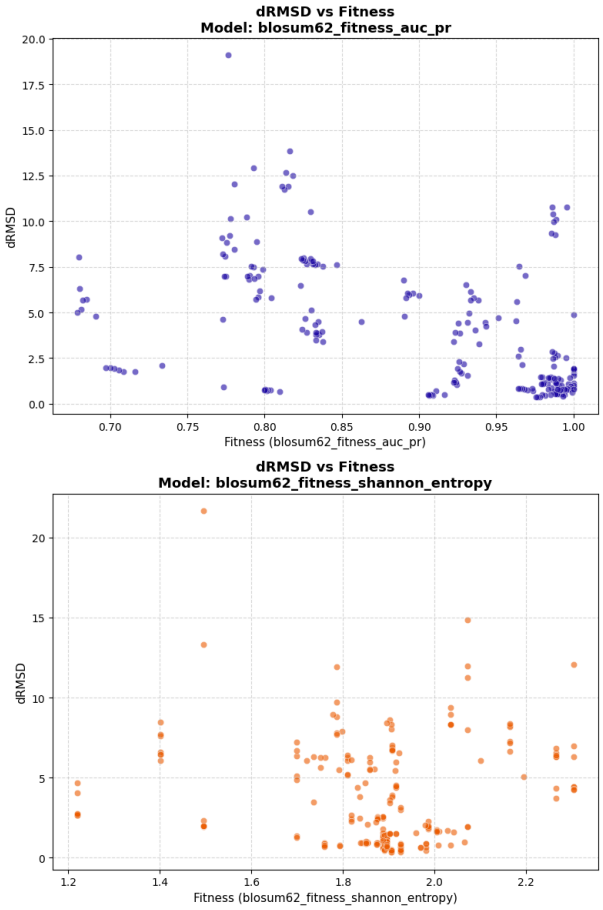


Figure 9: Relationship between fitness and dRMSD across evaluated individuals for both models.



A direct comparison between both heuristics did not reveal statistically significant differences in overall performance. However, under the best-case performance scenario, the Shannon entropy-based approach exhibited a slight advantage by achieving a statistically significant improvement over the random baseline in dRMSD, which was not observed for the AUC-PR-based method. Additionally, both models were capable of identifying near-native conformations for specific peptides, while consistently failing for others, suggesting that the effectiveness of these approaches is influenced by the intrinsic complexity of the target structures.

6.2 Analytical and Critical Reflection

Despite their effectiveness as exploratory optimization strategies, the results reveal important limitations in the use of contact maps as proxies for structural quality. In particular, sequences with lower fitness values were observed to yield better predicted three-dimensional structures than those with higher fitness, indicating that contact maps do not fully capture the complexity of structural similarity.

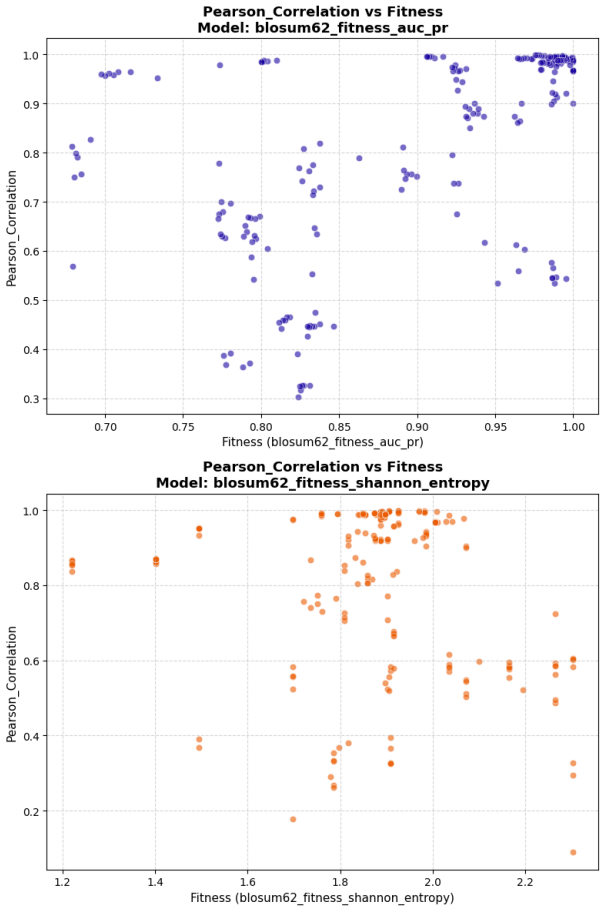
This limitation is further supported by the Spearman correlation analysis, which revealed a weak and inconsistent relationship between fitness values and structural performance metrics. Moreover, the top-ranked individual according to the fitness function corresponds to the best structural solution only in a limited proportion of cases, highlighting a clear mismatch between the optimization objective and true structural quality.

Overall, these findings indicate that contact map-based fitness functions are effective for guiding the search process but are not sufficiently reliable as final selection criteria. Their primary utility lies in enabling the identification of promising regions within the sequence space, which must subsequently be refined or validated using higher-resolution structural prediction methods.

6.3 Limitations

One important limitation of this study lies in the construction of the initial population. Given the extremely large search space, more robust strategies for population initialization should be explored. For the smallest peptide considered (15 residues), the search space is on the order of 20^{15}, while for the largest (30 residues), it increases to 20^{30}. Under these conditions, the quality and diversity of the initial population become critical factors influencing the effectiveness of the optimization process.

Figure 10: Relationship between fitness and Pearson correlation across evaluated individuals for both models.



Additionally, the potential impact of sequence length on the predictive performance of the models was not explicitly analyzed. Further investigation is required to determine whether peptide length systematically affects the quality of the structural predictions obtained through these approaches.

Additionally, this study considered only contact maps predicted by the ESM-2 model as the guiding signal for the optimization process. While ESM-2 provides a strong baseline, other contact map predictors could exhibit different behaviors in terms of accuracy and calibration of contact probabilities. Evaluating alternative predictors may reveal whether the effectiveness of the heuristic search is dependent on the underlying model, and whether improved or complementary predictors can provide stronger guidance toward structurally accurate conformations.

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